Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 . This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	A technology that enables network-resilient XR-based remote robot operation, allowing one operator to embody/supervise multiple robots across distributed sites to perform hazardous maintenance (e.g., nuclear) and daily industrial workflows (installation, programming, inspection, repair).			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1)]	Technology Field 1 (Large Category)		Technology Field 2 (Large Category)	
	Mechanical, Materials		Information & Communication	
	Medium Category	Small Category	Medium Category	Small Category
	Robotics/Automation Machinery	*Robot Control and Intelligence Technologies*	Digital Content	Virtual Reality, Computer Graphics
Concept of Proposed Technology	This proposed technology enables safe and intuitive XR-based remote teleoperation for robots operating across distributed sites, including hazardous environments and everyday industrial settings. Through an XR interface (AR/VR/MR) tightly coupled with a digital-twin representation, an operator gains high situational awareness and can embody, teleoperate, and supervise robots remotely, including switching between multiple robots and coordinating distributed robot			

	operations. The system provides four core capabilities: ● **Intuitive, user-centered XR control platform and interaction design:** The system provides a task-centric XR UI/UX that supports fast comprehension and low cognitive load, including (i) robot-state and safety-state visualization (ii) digital-twin aligned overlays for spatial awareness (iii) streamlined workflows for common tasks (programming/inspection/repair), and (iv) personalized assistance features such as adaptive guidance, skill-aware training, and actionable alerts to reduce operator burden and improve reliability. ● **Always-on safety assurance layer:** A safety controller continuously enforces safe operation through collision avoidance, virtual fixtures, safe-speed and force limits, and emergency fallback behaviors, ensuring stable operation even under unexpected operator input or environmental uncertainty. ● **Network-resilient teleoperation and stability control:** The control pipeline remains stable under delay, jitter, and packet loss using techniques such as predictive display, command smoothing, and local safe fallback autonomy, maintaining controllability and preventing unsafe robot actions during network degradation. ● **Multi-robot supervision and scalable operation (One operator, many robots)** A single operator can switch between robots across sites (and optionally supervise multiple robots concurrently), enabling scalable deployment aligned with a Robot-as-a-Service (RaaS) model for industrial maintenance, inspection, and intervention workflows. Target applications include hazardous-site intervention (e.g., nuclear facilities, confined spaces, high-risk maintenance) as well as routine industrial workflows such as robot programming, inspection, troubleshooting, and repair under real operational constraints.
Rationale for Joint R&D Project	A joint R&D project is essential because successful XR-based remote teleoperation requires the convergence of: ● human-centered XR/AI interaction and shared autonomy ● robust networked robot operation under real-world constraints (delay, jitter, packet loss, cybersecurity, and safety requirements) ● real industrial validation and system integration **Role of the Research Institute:** The research institute (Georgia Institute of Technology) will develop the core technologies, including the XR interaction framework and digital-twin interface, safety-aware shared autonomy (collision avoidance, virtual fixtures, safe-speed limits, emergency fallback), and operator modeling / learning-aware personalized assistance. The institute will also lead rigorous evaluation of workload, task performance, and safety using reproducible experimental protocols and human-factors metrics.

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	Role of the Korean SME Partner: The Korean SME partner will provide target use-cases, robot platforms, and operational constraints, and will supply field testing environments representative of deployment conditions. The SME will lead system integration, pilot deployment, and productization planning, including maintainability, onboarding workflows, and service operations considerations. This collaboration is required to ensure that the resulting system is not only technically innovative, but also deployable and scalable under real network conditions, safety standards, and industrial workflows / settings.
Key Words	• XR-based Remote Teleoperation • Network-Resilient Teleoperation Control • Safety-Critical Shared Autonomy • Multi-Robot Supervision & Fleet Teleoperation • Robot-as-a-Service (RaaS) / Remote Operations Platform

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL) [2)]	**Start of the Project**	Level (3)	**End of the Project**	Level (6)
R&D Objectives	The objective of this R&D project is to develop a safe, network-resilient XR-based remote teleoperation system that enables reliable robot operation across distributed industrial sites, by integrating XR digital-twin interfaces, safety-critical shared autonomy, robust networked control, and multi-robot supervision to support both hazardous-site intervention and routine industrial workflows. The specific R&D objectives are: ● To develop an XR teleoperation interface and digital-twin interaction framework that provides immersive situational awareness and task-centric UI/UX for industrial programming, inspection, troubleshooting, and repair. ● To develop a safety-critical shared autonomy layer that continuously enforces operational safety through collision avoidance, virtual fixtures, safe-speed/force limits, and emergency recovery behaviors under uncertain environments and operator inputs. ● To develop network-resilient teleoperation control and streaming methods that maintain stable operation under delay, jitter, and packet loss using predictive display, command smoothing, and robot-side safe fallback autonomy. ● To develop a multi-robot supervision and embodiment framework that allows a single operator to rapidly switch between robots/sites with preserved context, enabling scalable fleet operation aligned with a Robot-as-a-Service (RaaS) deployment model.			
R&D Target Specifications	**1. Safety** ● Safety events (near-collision / constraint violations): ≥ 50% reduction vs. baseline teleoperation. → Near-collision = minimum robot–obstacle separation < 10 cm, → Constraint violation = entering forbidden zone / exceeding speed limit / fixture violation. ● Robot-side safety override / E-stop initiation latency: ≤ 200 ms (trigger → safety-rated stop initiated / controller acknowledges safe-stop) **2. Network Robustness** ● Maintain stable teleoperation performance and task success maintained under influences of -RTT (round-trip time) up to 200 ms -Jitter up to 50 ms -Packet loss 2–5% *(random + burst loss profiles)* ● Automatic safe fallback activation: ≤ 500 ms after detecting severe degradation/disconnect **3. Operator Efficiency & Learning** ● Training time to reach defined proficiency (novice): ≥ 30% reduction ● Operator workload (NASA-TLX or equivalent): ≥ 15% reduction vs. baseline ● Task completion time (programming/inspection/repair benchmark set): ≥ 15% improvement vs. baseline, while maintaining task success/quality thresholds **4. Scalability** ● Fleet supervision capacity: single operator manages ≥ 3 robots via supervision + rapid switching (sequential embodiment) without degradation in safety KPIs			

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	• Context-switch time (Robot A → Robot B): ≤ 10 s while preserving situational awareness (state, task context, alerts). **5. Task Performance & Quality** • Task success rate: ≥ 90% on benchmark tasks. • Task quality metric: placement/inspection error ≤ 10% *(task-specific threshold defined with SME).* **6. Reliability & Recovery** • System availability during pilot study: ≥ 80% • Reconnect + resume time after temporary disconnect: ≤ 30 s
R&D Plan	This R&D project will be conducted over three years, following a progressive development and validation strategy from simulation to laboratory environments and finally to real-world domestic settings. Year 1 • Develop an initial XR-based teleoperation prototype to operate the laboratory robots, including digital-twin visualization and core operator controls. • Define benchmark tasks and measurable success criteria (task success, completion time, quality/error thresholds, and safety event definitions) to establish a clear baseline for comparison. • Design and evaluate the network and data streaming architecture (robot state, sensor/video streams, telemetry) under controllable delay, jitter, and packet-loss conditions. • Align deployment targets with the Korean SME partner by selecting candidate robot platforms and deciding on representative operational scenarios and constraints. Year 2 • Integrate the end-to-end system for safe remote teleoperation, including safety constraints, monitoring, and robot-side fallback behaviors for network degradation or disconnect. • Validate performance against the established benchmarks and demonstrate stable operation in representative laboratory scenarios, including reliability and recovery procedures. • Implement multi-robot supervision foundations (rapid switching and context preservation) and demonstrate feasibility in controlled scenarios. • Refine system integration with the SME platform by incorporating partner feedback on workflows, maintainability, and operational constraints, and update evaluation protocols accordingly. Year 3 • Conduct pilot-oriented validation to confirm stability, usability, and operator efficiency in representative industrial settings, with KPI-based reporting and documented results. • Perform structured operator evaluations (e.g., novice vs. experienced) to quantify training time, workload, and task efficiency improvements under realistic operating conditions. • Harden the system for deployment and prepare technology transfer and productization materials with the SME partner. • Finalize deployment packages (integration/operation guides, safety procedures, monitoring dashboards) and complete a pilot readiness review based on observed failure modes and recovery performance.

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	• Design and develop XR teleoperation UI/UX and digital-twin interaction • Develop safety-critical shared autonomy algorithms (virtual fixtures, collision avoidance, recovery) • Develop network-resilient teleoperation methods (predictive display, command smoothing, safe fallback) • Develop operator modeling and learning-aware assistance (skill estimation, adaptive feedback/training) • Conduct human-subject and performance evaluations (workload, task success, task completion time, user-studies). • Provide research infrastructure (protocols, data logging/analysis pipeline, documentation)
Expected R&R of Korean SME Partner	• Provide target industrial use-cases, workflow requirements, and success criteria. • Provide robot platform access (hardware/API) and deployment constraints. Support integration with safety interfaces. • Lead field system integration (robot, sensors, networking, operator station). • Provide test environments and participate in validation (pilot tasks, reliability, safety checks), including real network-condition testing. • Support pilot-scale prototype deployment planning, including site readiness, integration requirements, and operation/maintenance procedures. • Commit resources (engineering personnel, test time, pilot site access, hardware units).

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	**1) Technological Achievements** ● XR remote teleoperation platform validated in relevant environments, integrating: ○ an always-on safety-critical shared autonomy layer (collision avoidance, virtual fixtures, safe-speed limits, recovery/E-stop), ○ network-resilient teleoperation (delay/jitter/packet-loss tolerance with predictive display, command smoothing, robot-side fallback), and ○ multi-robot supervision and rapid embodiment switching across distributed sites. ● Learning-aware operator assistance that measurably improves usability, including skill estimation, adaptive training, and just-in-time guidance, delivering quantifiable reductions in training time and operator workload. ● Field-tested prototype (TRL 6) demonstrated in a pilot environment with operational logging/monitoring, reliability validation, and deployment feasibility under real workflow and network constraints. **2) Commercialization Potential** ● Scalable Robot-as-a-Service (RaaS) deployment model: centralized operators provide on-demand intervention and supervision for multiple robots across multiple customer sites, improving unit economics and service scalability. **3) Broader Industrial / Economic Impact** ● Reduced human exposure to high-risk environments (e.g., nuclear facilities, decommissioning sites, confined and cluttered industrial spaces) by enabling safe remote intervention. ● Improved SME productivity and service efficiency by enabling remote programming/inspection/repair, reducing dependence on scarce on-site experts, minimizing downtime, and accelerating troubleshooting cycles. ● Faster robot deployment and maintenance cycles, lowering operational barriers and supporting wider adoption of advanced robotics and intelligent automation across industrial sectors.

6. References

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization